

## Project Proposal

# UTHAO

end to end parcel delivery platform

# Team Members



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# Problem Statement

Manual courier systems break down at scale. Common failures:



**Parcel loss**



**Incorrect delivery status**



**Hard to track movement**



**Duplicate customer data**



**Poor warehouse management**



**No performance reports**

# Project Overview



## Customers

- Create delivery requests
- Track parcels
- View delivery history



## Branch Employees

- Receive parcels
- Assign to warehouses
- Update parcel status



## Delivery Agents

- Receive assignments
- Update delivery status



## Managers

- Monitor operations
- Generate reports
- Analyze performance

# Main Features



## User Management

Registration & role-based access



## Parcel Management

Create, update, cancel, track



## Warehouse Management

Add, store, transfer parcels



## Delivery Management

Assign, schedule, confirm



## Payment Management

Charges, history, revenue



## Reporting

Deliveries, revenue, performance

# Role of RDB



## Data Integrity

Primary keys, foreign keys & constraints keep data valid



## Data Consistency

Transactions keep assignments, payments & status in sync



## Efficient Querying

Joins, indexes & views for fast reporting



## Automation

Triggers & procedures run logic inside the database

# Core Entities

**Customer**

Sender & receiver info

**Branch**

Branch offices

**Warehouse**

Storage locations

**Employee**

Branch staff

**Delivery Agent**

Delivery personnel

**Parcel**

Parcel details & status

**Parcel Assignment**

Agent ↔ parcel link

**Status History**

Every parcel movement

**Payment**

Payment records

# PostgreSQL Features

Course-relevant techniques we will demonstrate



## Stored Procedures

Assign & transfer parcels



## Functions

Delivery cost, tracking history



## Triggers

Auto status history & revenue



## Views

Active & delayed deliveries



## Cursors

Row-by-row monthly reports



## Complex Joins

Multi-table analytics

# Tech Stack



**PostgreSQL**

Database



**Express.js**

Backend



**React**

Frontend



**Tailwind CSS**

Styling

# Workflow



# Advanced Features (Bonus)



## Materialized Views

Cached monthly revenue summary



## Partitioning

Status history split by month/year



## Audit Log Triggers

Track every update & delete



## Full-Text Search

Search by name or tracking number



## Transaction Mgmt

Atomic assign with rollback on fail

# Thank You

Questions?